

Frontend - Assignment 1

Q1. Explain the HTTP Request–Response Cycle with a suitable diagram. Describe the major steps involved when a client (browser) requests a resource from a web server and receives a response.

Answer

HTTP stands for hyper text transfer protocol. It is the process in which a browser sends a request to a web server and gets a response.

These are the following steps:-

- 1) User
- 2) Browser
- 3) HTTP Request
- 4) Web Server
- 5) HTTP Response
- 6) Web Page

The user enters a URL in the browser. The browser sends a request to the server. The server processes the request and sends the required webpage as a response. The browser then displays the webpage.

Q2. Briefly explain the purpose and key features of the following web technologies:

1. HTML
2. CSS
3. JavaScript
4. Bootstrap
5. jQuery
6. React.js

Answer

1. HTML

HTML stands for Hypertext Markup Language. It is used to make the basic structure of a webpage like headings, paragraphs, images, links and tables.

2. CSS

CSS stands for Cascading Style Sheets. It is used to style a webpage such as colors, fonts, spacing and layout.

3. JavaScript

JavaScript is used to make webpages interactive. It can handle clicks, change content and perform different actions.

4. Bootstrap

Bootstrap is a frontend framework which provides ready-made styles and components. It is also useful for making responsive websites.

5. jQuery

jQuery is a JavaScript library which makes tasks like changing HTML elements, events and animations easier.

Q3. What are client-side technologies and server-side technologies? Explain the difference between them with suitable examples.

Answer

Client-side technologies mainly run in the user browser. They are used for the visible part of the website and user interaction. Examples are HTML, CSS and JavaScript.

Server-side technologies run on the web server. They are used for processing requests, handling data and backend work. Examples are Node.js, PHP and Python.

Client-side runs in the browser, while server-side runs on the server.

Q4. What are block-level elements in HTML? Explain their characteristics and provide at least three examples.

Answer

Block-level elements that normally start from a new line and occupy the available width of the container.

Examples are <div>, <p>, <h1>, <section> .

Q5. What are semantic elements in HTML? Explain their importance in web development and provide suitable examples.

Answer

Semantic elements are HTML elements whose names tell about the content inside them. They make the structure of a webpage easier to understand.

Examples are <header>, <nav>, <section> and <footer>.

They also help search engines and screen readers understand the webpage better.

Q6. Write HTML code to create the following table:

Sr. No. Name Designation Department

1 John Manager Sales

2 Smith Sr. Manager Marketing

3 Jane Manager HR

The table should include appropriate table headings, rows, and columns.

Answer

```
<!DOCTYPE html>
<html>
<head></head>
<body>
```

```

<table border="3">
<tr>
<th>Sr. No.</th>
<th>Name</th>
<th>Designation</th>
<th>Department</th>
</tr>

<tr>
<td>1</td>
<td>John</td>
<td>Manager</td>
<td>Sales</td>
</tr>

<tr>
<td>2</td>
<td>Smith</td>
<td>Manager</td>
<td>Marketing</td>
</tr>

<tr>
<td>3</td>
<td>Jane</td>
<td>Manager</td>
<td>HR</td>
</tr>
</table>

</body>
</html>

```

Q7. Write HTML code to create the following list using an appropriate combination of ordered/unordered lists:

River

Ganga

Yamuna

Brahmaputra

Narmada

Answer

```

<!DOCTYPE html>
<html>
<head></head>
<body>

<ul>
<li>River
<ul>
<li>Ganga</li>
<li>Yamuna</li>
<li>Brahmaputra</li>

```

```
<li>Narmada</li>
</ul>
</li>
</ul>

</body>
</html>
```

Q8. Explain the different types of CSS used for applying styles to an HTML document. Describe each type with a suitable example.

Answer

There are three types of CSS.

1. Inline CSS

It is written directly inside an HTML tag.

```
<p style="color:red;">Hello</p>
```

it can affect only line in which it is used.

2. Internal CSS

It is written inside the style tag in the same HTML file.

it affects only one specific html page .

3. External CSS

It is written in a separate CSS file.

It can affect all Html pages linked to it.

Q9. Differentiate between HTML, CSS, and JavaScript based on their role in developing a web page. Explain what each technology contributes to a website.

Answer

HTML is used to create the structure of a webpage.

CSS is used to design the webpage and control its appearance.

JavaScript is used to add interaction and make the webpage work dynamically.

HTML gives the structure, CSS gives the design and JavaScript gives the interaction.

Q10. Answer the following:

a. What is the role of Bootstrap in web development?

Answer

Bootstrap is a frontend framework which helps in creating responsive websites quickly. It provides ready-made components like buttons, forms and navigation bars.

b. What is jQuery and why was it widely used in web development?

Answer:

jQuery is a JavaScript library which makes JavaScript tasks easier. It was widely used for DOM manipulation, events, animations and AJAX.

c. What is React.js and what problem does it solve in frontend development?

Answer:

React.js is a JavaScript library used for building user interfaces. It uses reusable components, which makes large and interactive frontend applications easier to manage.